

UPDATE Basics

The UPDATE statement modifies existing rows in a table.

General syntax:

```
UPDATE table_name  
SET column_name = value;
```

Suppose we execute

```
UPDATE Students  
SET Age = 22;
```

MySQL performs:

- Read the table.
- Find all rows (no WHERE clause).
- Lock the rows (depending on storage engine and transaction isolation).
- Modify the values.
- Write changes to disk.
- Write to the transaction log (redo/undo logs).

Updating a Single Column

Example:

```
UPDATE Students  
SET Age = 21  
WHERE StudentID = 1;
```

Updating Multiple Columns

Separate columns using commas.

```
UPDATE Students
SET
Age = 21,
Department = 'ECE'
WHERE StudentID = 1;
```

Multiple conditions

```
UPDATE Students
SET Department = 'ECE'
WHERE Age > 20
AND CGPA > 8;
```

```
UPDATE Students
SET Department = 'ECE'
WHERE StudentID IN (1, 3, 5);
```

```
UPDATE Employees
SET Salary = Salary + 5000
WHERE Salary BETWEEN 40000 AND 60000;
```

```
UPDATE Employees
SET Salary = 100000;
```

Every employee now has the same salary.
This is one of the most common production mistakes.

Updating Using CASE

Useful for conditional updates.

Example

```
UPDATE Employees
SET Salary =
CASE
    WHEN Salary < 50000 THEN Salary * 1.20
    WHEN Salary < 100000 THEN Salary * 1.10
    ELSE Salary * 1.05
END;
```

```
UPDATE Students
SET Grade =
CASE
    WHEN CGPA >= 9 THEN 'A'
    WHEN CGPA >= 8 THEN 'B'
    ELSE 'C'
END;
```

LC 627 — Swap Salary

Table

id	name	sex
1	Aadya	f
2	Arpit	m

Need to swap m ↔ f.

Solution

```
UPDATE Salary
SET sex =
CASE
    WHEN sex = 'm' THEN 'f'
    ELSE 'm'
END;
```

Updating NULL Values

Suppose

Name	Phone
Aadya	NULL

Update missing phone numbers

```
UPDATE Students
SET Phone = '9999999999'
WHERE Phone IS NULL;
```

```
UPDATE Students
SET Age = 20;
```

You may see:

Error Code: 1175. You are using safe update mode...

Safe Update Mode prevents updates that:

- Don't use a WHERE clause with a key column.
- Or don't use LIMIT.

To temporarily disable it:

```
SET SQL_SAFE_UPDATES = 0;
```

Re-enable:

```
SET SQL_SAFE_UPDATES = 1;
```

Difference between UPDATE and ALTER?

UPDATE changes data.

ALTER changes the table structure.

The Scenario: You have two tables: Users and Banned_IPs. You need to lock the accounts of any user who logged in from a banned IP address.

Concepts used: UPDATE, SET, WHERE, IN (Subquery)

The Solution:

SQL

```
UPDATE Users
SET is_locked = True
WHERE ip_address IN (
    SELECT ip_address
    FROM Banned_IPs
);
```